

Phase 1 — Finish the binary classification (highest priority)

1. Complete the comparison section

Create one final comparison table.

Model	Destination Port	Accuracy	Precision	Recall	F1	Macro F1	Train Time
Decision Tree	✓						
Decision Tree	✗						
Random Forest	✓						
Random Forest	✗						
XGBoost	✓						
XGBoost	✗						

Then conclude:

- Which model is best?
 - Does removing Destination Port significantly affect performance?
-

2. Add feature importance

For:

- Decision Tree
- Random Forest
- XGBoost

Show:

- Top 10 features
 - Explain why Destination Port is (or isn't) important.
-

3. Save the final binary model

models/

best_binary_model.pkl

This is what your teammate will integrate into Streamlit.

Phase 2 — Multi-class classification

Once the binary pipeline is complete:

Restore the original labels.

Instead of:

BENIGN

ATTACK

use:

BENIGN

DDoS

DoS Hulk

PortScan

...

Heartbleed

Repeat the experiments:

- Decision Tree
- Random Forest
- XGBoost

No need to test with/without Destination Port again unless you specifically want to study its effect in the multiclass setting.

Phase 3 — Deep Learning (optional but valuable)

Only after the classical ML models are finished.

Start with one model:

- **MLP (Multi-Layer Perceptron)**

If you have enough time, add:

- 1D CNN **or**
- LSTM

Then compare:

Model	Type	Accuracy	Macro F1
Decision Tree	ML		
Random Forest	ML		
XGBoost	ML		
MLP	Deep Learning		

This gives your project an extra contribution without becoming too large.

Phase 4 — Final integration

Once the best model is selected:

- Export it with joblib
- Your teammate connects it to Streamlit
- Test predictions from uploaded CSV files
- Generate charts and summaries

Phase 5 — Final report

Complete:

- Results chapter
- Discussion
- Conclusion
- Future work